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Effectiveness of manual therapy, exercise, low-level laser therapy, ultrasound, and transcutaneous electrical nerve stimulation in reducing pain and improving mouth opening in temporomandibular joint disorders

A network meta-analysis of randomized controlled trials

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Abstract

Background: Temporomandibular joint (TMJ) disorders are prevalent musculoskeletal conditions. Common therapies include TMJ manual therapy, TMJ exercise, and modalities such as low-level laser, therapeutic ultrasound, and transcutaneous electrical nerve stimulation (TENS). This study ranked TMJ manual therapy, TMJ exercise, low-level laser therapy, therapeutic ultrasound, and TENS for pain relief and improved maximal mouth opening (MMO) to strengthen evidence for clinical practice.

Methods: PubMed, Scopus and Ovid MEDLINE were systematically searched from inception to March 2026 for randomized controlled trials evaluating TMJ manual therapy, TMJ exercise, low-level laser therapy, therapeutic ultrasound, and TENS for TMJ disorder. A frequentist network meta-analysis with a random-effects model was performed to compare treatment effects across interventions, with standardized mean differences calculated as the effect size. Treatment rankings were estimated using the surface under the cumulative ranking curve. Publication bias was assessed using funnel plots and Egger test.

Results: This network meta-analysis demonstrated significant pain reduction with therapeutic ultrasound, TMJ manual therapy, TMJ exercise, and low-level laser therapy, and TENS. Therapeutic ultrasound showed the highest probability of providing pain relief, followed by TMJ manual therapy, low-level laser therapy, TMJ exercise, and TENS. For MMO, TMJ manual therapy, low-level laser therapy, therapeutic ultrasound, and TENS showed significant improvements, while TMJ exercise demonstrated a borderline effect. Based on the surface under the cumulative ranking curve rankings, TMJ manual therapy had the highest probability ranking, followed by therapeutic ultrasound.

Conclusion: This network meta-analysis suggests that TMJ manual therapy had a higher probability ranking than TMJ exercise for reducing pain and improving MMO. Among physical modalities, therapeutic ultrasound and low-level laser therapy tended to rank higher than TENS. These findings may help inform clinical decision-making in the management of TMJ disorder. Future studies should further investigate the influence of treatment duration and TMJ disorder diagnostic subtypes.

Abbreviations: CI = confidence interval, MMO = maximal mouth opening, NRS = Numeric Rating Scale, PRISMA = Preferred Reporting Items for Systematic Reviews and Meta-Analyses, RCTs = randomized controlled trials, SMD = standardized mean difference, SUCRA = surface under the cumulative ranking curve, TENS = transcutaneous electrical nerve stimulation, TMJ = temporomandibular joint, VAS = Visual Analog Scale.

Keywords: chewing, laser, mouth, pain, rehabilitation, TENS, ultrasound

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All data generated or analyzed during this study are included in this published article [and its supplementary information files].

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1. Introduction

Temporomandibular joint (TMJ) disorders are prevalent musculoskeletal conditions that can lead to pain, restricted jaw function. The overall prevalence of TMJ disorders ranges from 21.5% to 50.5% and is more common in females. TMJ disorders are a major cause of non-odontogenic orofacial pain, with 40% to 75% of individuals exhibiting at least one clinical sign (e.g., TMJ noise) and approximately 33% reporting symptoms such as TMJ or facial pain.[1] Symptoms vary widely from mild pain and temporary jaw discomfort to severe chronic conditions characterized by intense pain and substantial limitations in jaw function, such as restricted mouth opening.[2] TMJ disorders can be classified into articular and nonarticular categories. The latter presents as myofascial pain in the chewing muscles, whereas articular disorders involve internal derangements characterized by alterations in the relationship between the disc and condyle, such as disc displacement.[3] The Research Diagnostic Criteria for TMJ disorders classification system is widely used in research and clinical settings. It comprises 2 axes – Axis I addresses physical diagnoses, and Axis II evaluates psychosocial factors. Axis I is further divided into 3 categories – muscle disorders (Group I), disc disorders (Group II), and arthralgia (Group III).[4]

Several conservative treatments are available for TMJ disorders, including physical therapy, medications (e.g., anti-inflammatory drugs, analgesics, and muscle relaxants), removable oral splints, and acupuncture.[5] Common physical therapy approaches include TMJ manual therapy, TMJ exercise, and physical modalities. TMJ manual therapies, such as joint mobilization, manipulation, and soft tissue treatment, and TMJ exercise, such as mouth opening exercise, are frequently used. A systematic review and meta-analysis have confirmed the effectiveness of both methods in reducing pain intensity.[6] TMJ manual therapy may restore the range of motion, reduce local ischemia, stimulate proprioception, break adhesions, enhance synovial fluid production, and alleviate pain. Furthermore, TMJ exercise may improve strength, mobility, and coordination while relieving joint and muscle pain.[6] Most review articles on physical modalities for TMJ disorders management have focused on low-level laser therapy,[1] transcutaneous electric nerve stimulation (TENS),[7] and therapeutic ultrasound.[8] These modalities are widely used because of their potential analgesic, anti-inflammatory, and muscle-relaxing effects.[9]

Several network meta-analyses have investigated conservative treatments for TMJ disorders,[3,10,11] but none have ranked individual physical therapy interventions. Such an analysis would enable a clearer comparison of their relative effectiveness and aid in the development of clinical guidelines. Accordingly, this network meta-analysis aimed to rank the efficacy of TMJ manual therapy, TMJ exercise, and physical modalities in reducing pain intensity and improving maximal mouth opening (MMO) while focusing on their independent effects. This approach provides a precise evaluation of the specific efficacy of each intervention while minimizing potential confounding factors.

2. Methods

2.1. Search strategy

This study was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) extension guidelines for network meta-analysis.[12] The PRISMA checklist is included in Table S1, Supplemental Digital Content 1. The study was registered with INPLASY under the registration number (INPLASY202520007). Ethical review board approval and informed consent of the participants were not required for this study. Ethical approval was not required because this study was based on previously published studies and did not involve human participants or identifiable data.

Two authors (L-HL and T-YL) independently conducted electronic searches in PubMed, Scopus and Ovid MEDLINE using the following keywords: "therapeutic ultrasound" OR "low level laser" OR "transcutaneous electrical nerve stimulation" OR "manual therapy" OR "exercise" AND "temporomandibular joint disorders" OR "temporomandibular joint dysfunction." The search spanned from the earliest available records in each database to the most recent date (March 2026). Titles and abstracts of the identified studies were screened for eligibility by the 2 authors using a consensus-based approach. Disagreements were resolved through consultation with the corresponding authors. No language restrictions were imposed. Further details of the search strategy are provided in Table S2, Supplemental Digital Content 2.

2.2. Inclusion criteria

The Population, Intervention, Comparison, and Outcome framework for this meta-analysis was defined as follows: the population comprised adults (≥18 years) diagnosed with TMJ disorders; the interventions included TMJ manual therapy, TMJ exercise, low-level laser therapy, therapeutic ultrasound, and TENS; the comparator consisted of no intervention or sham therapy; and the outcomes were changes in pain intensity and MMO.

The inclusion criteria were as follows: randomized controlled trials (RCTs) investigating pain intensity and/or MMO,[2] studies involving adults diagnosed with TMJ disorders based on medical imaging or the Research Diagnostic Criteria for TMJ disorders classification system (Axis I), and intervention groups receiving a single treatment.

2.3. Exclusion criteria

The exclusion criteria were as follows: non-RCTs; patients with a history of surgery, inflammatory arthritis, fractures, dislocations, or burns; studies lacking assessments of pain intensity or MMO; treatments combined with different types of physical therapies or other conservative treatments (e.g., medication injections, oral nonsteroidal anti-inflammatory drugs, acupuncture, or splint treatments); comparative groups including injections, oral nonsteroidal anti-inflammatory drugs, acupuncture, or splint treatments; all included groups receiving similar treatments; age < 18 years old; and treatment duration only 1 day. Similar treatments were defined as interventions that could reasonably be classified within the same treatment node based on shared therapeutic mechanisms and clinical intent. For example, low-level laser therapies delivered at different power outputs, therapeutic ultrasound administered at different frequencies, various manual therapy techniques, and different forms

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of exercise therapy were considered similar treatments and grouped within the same node.

2.4. Modeling for network meta-analysis

Intervention classification was determined according to the specific components of the treatment protocols reported in each study. Two reviewers (L-HL and T-YL) independently evaluated and categorized the interventions. Any disagreements were discussed and resolved through consultation with the corresponding author (K-VC). For the purposes of the network meta-analysis, the interventions were grouped into 6 treatment nodes: TMJ manual therapy,[2] TMJ exercise,[3] low-level laser therapy (referred to as low level laser in the figures and tables),[4] therapeutic ultrasound,[5] TENS, and control treatment. Manual therapy was defined as interventions involving joint mobilization, manipulation, or soft tissue treatment.[13] TMJ exercise was defined as therapeutic exercises targeting the TMJ, including active mouth-opening exercise, passive mouth-openingexercise, TMJ stretching exercise, TMJ resistance exercise, jaw relaxation exercises, and Rocabado's 6 × 6 exercise program.[14] For therapeutic ultrasound, low-level laser therapy, and TENS, interventions were not further categorized according to specific modality parameters (e.g., pulsed vs continuous mode). Further stratification would have substantially fragmented the network and produced disconnected nodes, thereby limiting the feasibility of meaningful comparative analyses. Furthermore, previous meta-analysesinvestigating the use of therapeutic ultrasound and low-level laser therapy in other musculoskeletal conditions have reported that differences in treatment modes (e.g., continuous vs pulsed delivery) did not result in statistically significant differences in clinical outcomes.[15,16] Interventions involving no active therapy, sham procedures or education were categorized as control treatments.

2.5. Methodological quality appraisal

The Physiotherapy Evidence Database scale was used to assess the quality of the RCTs included in this meta-analysis. This scale evaluates 11 key criteria to determine study quality, including eligibility criteria, random allocation, allocation concealment, baseline comparability, participant blinding, therapist blinding, assessor blinding, outcome measurements, intention-to-treat analysis, between-group comparisons, and reporting of point estimates and variability measures.[17] Each criterion is scored from 0 to 10, with the eligibility criterion excluded from the total score. Studies scoring 6 or above were classified as good or excellent quality, those scoring between 4 and 5 as fair quality, and those scoring 3 or below as poor quality.

2.6. Primary outcome

This study primarily aimed to evaluate changes in pain intensity using the Numeric Rating Scale (NRS) and Visual Analog Scale (VAS) before and after the intervention. The NRS, available as an 11-point (NRS-11, 0–10) or 101-point (NRS-101, 0–100) scale, is widely used in clinical settings because of its simple administration and scoring. The VAS requires patients to represent their pain as a point along a 10 cm line on paper.[18] For the meta-analysis, the NRS-11 and 10-cm VAS were used to standardize measurements and calculate effect sizes.

2.7. Secondary outcome

The secondary outcome was the MMO before and after treatment. Participants, in their intercuspal position, were instructed to open their jaws. The distance between the maxillary and mandibular central incisors was measured using a ruler or Vernier calipers.

2.8. Data extraction

Two independent reviewers (L-HL and T-YL) performed extraction of data, including demographic information, study design, intervention details, outcome measures, and assessment timings. The reasons for exclusion were meticulously documented. In cases of disagreement, a consensus was reached, or the corresponding author was consulted for resolution. For studies with multiple posttreatment data points, outcomes from the end of the intervention were selected for analysis. In multi-arm studies, similar eligible groups were combined for network comparisons.[19] For example, sham low-level laser therapy and sham TENS were classified as the control group. If one group received a combination treatment while the others received single treatments, only data from the single-treatment groups were retained, and data from the combination-treatment group were excluded.[19] Missing data in published articles were addressed by contacting the corresponding authors for original data. After data acquisition, medians, 95% confidence intervals (CIs), and interquartile ranges were converted to means and standard deviations in accordance with the guidelines outlined in the Cochrane Handbook for Systematic Reviews of Interventions.[20]

2.9. Statistical analysis

A frequentist network meta-analysis was conducted using the *network meta* command in Stata Basic Edition (Stata Statistical Software: Release 19.5; StataCorp LLC), which is based on a multivariate meta-analysis framework implemented through the *mvmeta* procedure. A random-effects model assuming a common between-study variance (τ^2) across treatment comparisons was prespecified. Between-study heterogeneity was estimated using restricted maximum likelihood.[21] Multi-arm trials were entered as arm-level data using the network setup command in Stata. Correlations among effect estimates arising from the same multi-arm study were accounted for within the multivariate meta-analysis framework of the *network meta* command (implemented via *mvmeta*), thereby avoiding double-counting and preserving independence across studies.

Two independent reviewers (L-HL and T-YL) assessed the distribution of potential effect modifiers across treatment comparisons (Figs. S1 and S2, Supplemental Digital Content 3). Overall, most interventions showed comparable distributions in treatment duration and participant age, supporting the plausibility of the transitivity assumption. However, studies involving joint-related TMJ disorder were relatively limited, and several studies did not clearly report the specific TMJ disorders diagnostic category. To further evaluate the potential influence of these variables, meta-regression analyses were conducted to examine whether treatment duration, participant age and TMJ disorders diagnostic subtype moderated treatment effects across different interventions. TMJ disorders diagnoses were categorized into 4 groups: TMJ disorder-myogenous, TMJ disorder joint-related (including disc displacement or arthrogenous disorder), TMJ disorder-mixed (a combination of myogenous and joint-related conditions), and TMJ disorder -no description when the diagnostic subtype was not reported. Regarding participant age, when the overall mean age of participants was reported in the study, this value was directly extracted. If only group-specific mean ages were available, a pooled mean age was calculated across groups. When studies reported only an age range, the midpoint of the reported range was used to approximate the mean age for analysis.

We generated forest and network plots to visualize pairwise comparisons between individual studies. Forest plots were created to illustrate standardized mean differences (SMDs) in pain

intensity and MMO for each treatment type compared to the control treatment, summarizing the effects. SMDs was expressed as point estimates with corresponding 95% CIs. To aid clinical interpretation, treatments were ranked for each outcome using the surface under the cumulative ranking curve (SUCRA), which quantifies the cumulative probability that a treatment is superior to a theoretical comparator, reflecting its relative efficacy. Potential inconsistencies between direct and indirect evidence were assessed using the side-splitting method to identify discrepancies.[22] Statistical significance was defined as a two-tailed *P*-value of <.05.

Sensitivity analysis was used to assess the impact of the pre–post correlation coefficient. To transform baseline and post-intervention pain intensity and range data into the mean and standard deviation of changes, an assumed correlation coefficient between pre- and post-intervention values is necessary. This study used a coefficient of 0.8, as recommended by previous study.[23] Given that alternative values are commonly applied in the literature (e.g., 0.5 or 0.3),[23,24] additional sensitivity analyses using correlation coefficients of 0.5 and 0.3 were performed to assess the robustness of the results in this network meta-analysis.

On the other hand, to assess potential publication bias, we followed the guidelines outlined in the Cochrane Handbook for Systematic Reviews of Interventions. This included visually inspecting funnel plots and performing Egger regression test.[25]

3. Results

3.1. Literature search

A PRISMA flowchart for the literature search is shown in Figure 1. Of the 834 non-duplicated articles identified, 65 were further analyzed for eligibility. Fourty-three of them were excluded after full-text review, including 6 due to treatments involving oral medications or injections, 1 for being a non-RCT, 3 for lacking pain intensity and MMO assessments, 6 for missing calculable data on mean or standard deviation values, 6 for treatments involving splints in combination with other interventions, 4 for similar treatments in both the experimental and control groups, 2 for including controls without TMJ disorders, 2 for including participants who primarily underwent TMJ surgery or had cervicofacial burns, 7 for treatment with combined therapies, 4 for participant age < 18 years old, and 2 for treatment duration treatment duration of only 1 day. The specific reasons for exclusion are listed in Table S3, Supplemental Digital Content 4.

3.2. Study characteristics

We included 22 RCTs (952 participants; mean age, 18–50 years; intervention duration, and 1–8 weeks) consisting of 14 two-arm studies[26–39] and 8 multi-arm studies.[40–47] Moreover, 20 RCTs focused on pain intensity whereas 15 assessed MMO. Details regarding the mean age of participants, exercise intervention specifics, and outcome assessment scales are provided in Table 1. The nodes of the current network meta-analysis were categorized according to TMJ manual therapy, TMJ exercise, therapeutic ultrasound, low-level laser therapy, TENS, and control therapies (e.g., sham therapy, no treatment, or education). The network model for the interventions is shown in Figure 2A and B.

3.3. Methodological quality of the included studies

Regarding the methodological quality of the included studies, we observed that 1 RCT received a score of 9, 1 RCT received a score of 8, 4 RCTs received a score of 7, 3 RCTs received a score of 6, 2 RCTs received a score of 5, 10 RCTs received a score of 4, and 1 RCT received a score of 3. Most RCTs did not report whether participants were lost to follow-up or whether allocation concealment was implemented. The details are provided in Table S4, Supplemental Digital Content 5.

3.4. Primary outcome

Overall, TMJ manual therapy, low-level laser therapy, and therapeutic ultrasound showed significant improvements in pain intensity (TMJ manual therapy: SMD = −2.58, 95% CI = −4.56

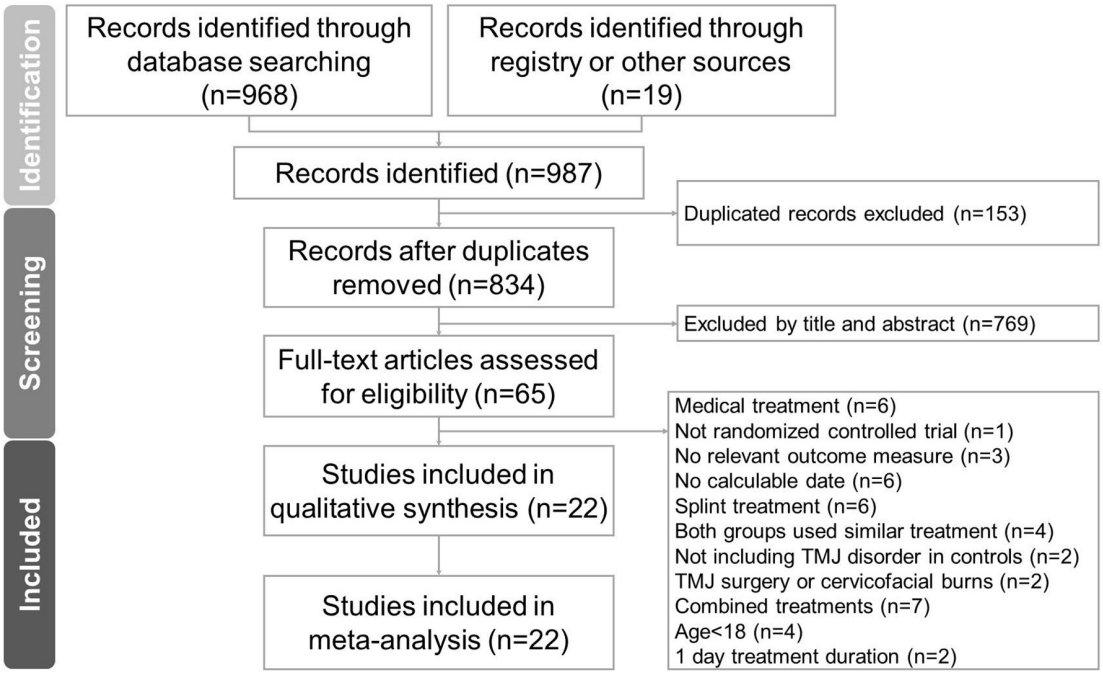


Figure 1. PRISMA flow diagram illustrating the screening and review process for the network meta-analysis. PRISMA = Preferred Reporting Items for Systematic Reviews and Meta-Analyses, TMJ = temporomandibular joint.

Table 1

Summary of the included trials investigating the effects of investigated interventions to reduce pain intensity and improve maximal mouth opening.

References	Study design	Enrolled population	Participants (N; F/M)	Age	Treatment frequency/duration	Outcome measure
Arabion et al[26]	RCT	TMJ disorders-mixed	TENS (30; 20/10) Control (30; 22/8)	TENS: 31.33 ± 8.26 Control: 28.73 ± 6.9	7 d with 7 sessions	VAS MMO
Bae and Park[40]	RCT	TMJ disorder-no-description	TMJ exercise (30; 23/7) Control (11; 8/3)	23.2 ± 6.9	3 times/d, 4 wks	VAS MMO
Barbosa et al[27]	RCT	TMJ disorders-mixed	TMJ exercise (17; 17/0) Control (17; 17/0)	TMJ exercise: 30 ± 7 Control: 26 ± 8	2 sessions/wk, 8 wks	VAS
Ba et al[28]	RCT	TMJ disorder-no description	Therapeutic ultrasound (80;* 37/43) Control (80;* 38/42)	Therapeutic ultrasound: 35.8 ± 9.7 Control: 36.0 ± 8.0	5 sessions/wk, 2 wks†	VAS MMO
Bijjaragi et al[29]	RCT	TMJ disorders-myogenous	TENS (20) Control (20)	20–60 yrs	Total 4 sessions, 1 wk	VAS MMO
Brochado et al[41]	RCT	TMJ disorders-mixed	TMJ manual therapy (13; 12/1) Low level laser (14; 14/0)	TMJ manual therapy: 41.2 ± 20.3 Low-level laser: 45.7 ± 15.7	3 sessions/wk, 4 wks	VAS MMO
Budakoti et al[42]	RCT	TMJ disorders-joint related	Low level laser (15) Therapeutic ultrasound (15) TENS (15)	20–50 yrs	2 sessions/wk, 4 wks	VAS MMO
da Cunha et al[30]	RCT	TMJ disorders-mixed	Low level laser (20) Control (20)	Low-level laser: 40.15 Control: 46.6	1 session/wk, 4 wks	VAS
De Abreu Venancio et al[39]	RCT	TMJ disorders-joint related	Low level laser (15; 13/2) Control (15; 12/3)	Low-level laser: 34.9 Control: 37.6	2 sessions/wk, 3 wks‡	VAS MMO
Emshoff et al[31]	RCT	TMJ disorder-no description	Low level laser (26)* Control (26)*	Low-level laser: 44.1 ± 16.6 Control: 41.8 ± 11.2	2–3 sessions/wk, 8 wks	VAS
Gavish et al[32]	RCT	TMJ disorders-myogenous	TMJ exercise (10) Control (10)	TMJ exercise: 27.1 ± 10.1 Control: 27.3 ± 5.9	3 times/d, 8 wks	VAS
Kalamir et al[33]	RCT	TMJ disorders-myogenous	TMJ manual therapy (23)* TMJ exercise (23)*	TMJ manual therapy: 28.2 ± 9.43 TMJ exercise: 26.8 ± 6.81	2 sessions/wk, 5 wks	VAS MMO
Kim[43]	RCT	TMJ disorders-mixed	Low level laser (12; 5/7) TENS (12; 5/7) Control (12; 7/5)	Low-level laser: 21.75 ± 2.86 TENS: 23.50 ± 2.07 Control: 21.58 ± 3.20	3 sessions/wk, 2 wks	MMO
Kirupa et al[34]	RCT	TMJ disorder-no description	TENS (15) Therapeutic ultrasound (15)	20–40 yrs	4 wks	VAS
Monteiro et al[35]	RCT	TMJ disorders-mixed	Low level laser (22; 17/5) Control (20; 15/5)	Low-level laser: 29.1 ± 11 Control: 25.6 ± 8	1 session/wk, 4 wks	VAS MMO
Patil and Aileni[36]	RCT	TMJ disorders-myogenous	TENS (18; 11/7) Control§ (18; 12/6)	TENS: 32.91 ± 12.57 Control: 34 ± 7.4	1 session/wk, 4 wks	MMO
Ramesh et al[37]	RCT	TMJ disorders-myogenous	TENS (15; 13/2) Therapeutic ultrasound (15; 12/3)	28.9 ± 6.02	7 sessions/wk, 1 wk	VAS
Sancakli et al[44]	RCT	TMJ disorders-myogenous	Low level laser (20) Control (10)	39.2 ± 2.8	3 sessions/wk, 4 wks	VAS MMO (no complete data)
Sarfraz et al[38]	RCT	TMJ disorders-myogenous	TMJ manual therapy (12) TMJ exercise (12)	18–55 yrs	3 sessions/wk, 6 wks	NPRS MMO
Seifi et al[45]	RCT	TMJ disorders-myogenous	Low level laser (10) Control (20) TENS (10)	18–50 yrs	4 wks	VAS MMO
Tanhan et al[46]	RCT	TMJ disorders-myogenous	Low level laser (17) TMJ exercise (20) TMJ manual therapy (22)	Low-level laser: 21.41 ± 4.30 TMJ exercise: 20.45 ± 1.50 TMJ manual therapy: 20.95 ± 1.83	3 sessions/wk, 4 wks	VAS MMO
Yeladandi et al[47]	RCT	TMJ disorders-joint related	TENS (10) Low level laser (10) Therapeutic ultrasound (10)	33.6	4 wks	VAS MMO

MMO = maximal mouth opening, NRS = Numeric Rating Scale, TENS = transcutaneous electrical nerve stimulation, TMJ = temporomandibular joint, VAS = Visual Analog Scale.

*Original allocated sample size.

†Outcome measured at 4 weeks.

‡Outcome measured on day 30.

§Although the intervention description in the original study mentioned lifestyle modification and home exercises, these were described only as advice without supervision. Therefore, this group was classified as a control node.

to – 0.60; low-level laser therapy: SMD = –2.04, 95% CI = –3.16 to – 0.93; therapeutic ultrasound: SMD = –2.74, 95% CI = –4.39 to – 1.08; TMJ exercise: SMD = –1.69, 95% CI = –3.32 to –0.07) and TENS showed a positive trend for pain improvement (SMD = –1.39, 95% CI = –2.28 to 0.03; Fig. 3).

The interventions were ranked based on their SUCRA, with followed by therapeutic ultrasound (mean rank 1.9), TMJ manual therapy (mean rank 2.1), low-level laser therapy (mean rank 3.1), TMJ exercise (mean rank 3.7), TENS (mean rank 4.3) and control (mean rank 5.9) in that order. Please see Table S5,

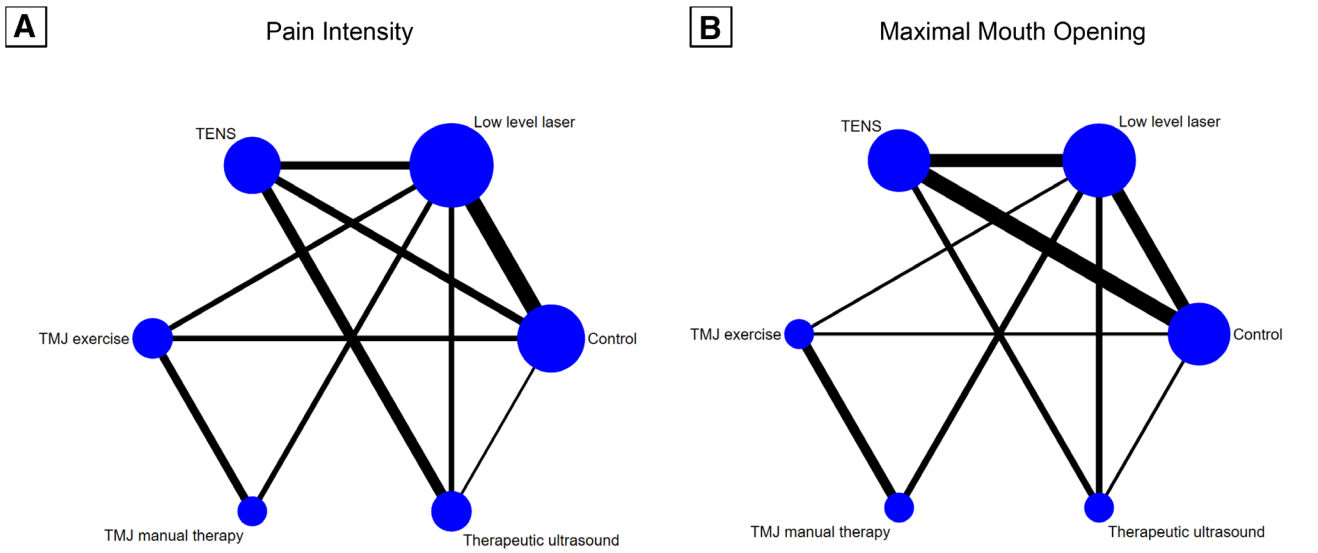


Figure 2. Network node plot of outcome measures for (A) pain intensity and (B) maximal mouth opening. TENS = transcutaneous electrical nerve stimulation, TMJ = temporomandibular joint.

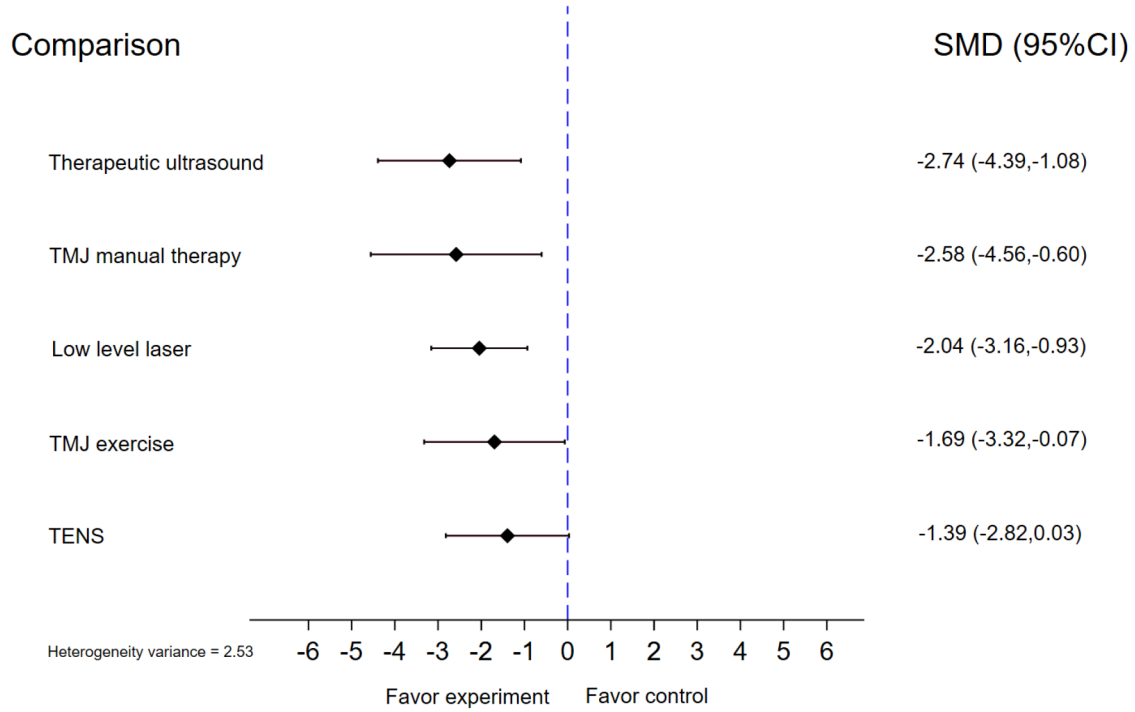


Figure 3. Forest plot of pain intensity comparing interventions with control. SMD = standardized mean difference, TENS = transcutaneous electrical nerve stimulation.

Table 2

League table of pain intensity.

Control	−2.04 (−3.16 to −0.93)	−1.39 (−2.82 to 0.03)	−1.69 (−3.32 to −0.07)	−2.58 (−4.56 to −0.60)	−2.74 (−4.39 to −1.08)
2.04 (0.93 to 3.16)	**Low level laser**	0.65 (−0.80 to 2.10)	0.35 (−1.21 to 1.91)	−0.54 (−2.35 to 1.28)	−0.69 (−2.34 to 0.95)
1.39 (−0.03 to 2.82)	−0.65 (−2.10 to 0.80)	**TENS**	−0.30 (−2.30 to 1.70)	−1.19 (−3.44 to 1.06)	−1.34 (−2.81 to 0.12)
1.69 (0.07 to 3.32)	−0.35 (−1.91 to 1.21)	0.30 (−1.70 to 2.30)	**TMJ exercise**	−0.89 (−2.54 to 0.77)	−1.04 (−3.20 to 1.11)
2.58 (0.60 to 4.56)	0.54 (−1.28 to 2.35)	1.19 (−1.06 to 3.44)	0.89 (−0.77 to 2.54)	**TMJ manual therapy**	−0.16 (−2.54 to 2.23)
2.74 (1.08 to 4.39)	0.69 (−0.95 to 2.34)	1.34 (−0.12 to 2.81)	1.04 (−1.11 to 3.20)	0.16 (−2.23 to 2.54)	**Therapeutic ultrasound**

TENS = transcutaneous electrical nerve stimulation, TMJ = temporomandibular joint.

Supplemental Digital Content 6 and Figure S3, Supplemental Digital Content 7 for a detailed SUCRA and ranking of the treatment types. The detailed pairwise comparison results for study arms and individual studies are presented in Figure S4A, Supplemental Digital Content 8 and Figure S4B, Supplemental Digital Content 9, as well as in Table 2.

In the sensitivity analyses, the assumed pre–post correlation coefficient was varied from 0.8 to 0.5 or 0.3, after which the network meta-analysis was repeated. The direction of SMD estimates, treatment rankings, and overall interpretations remained consistent with the main analysis (Fig. S5, Supplemental Digital Content 10 and Tables S6 and S7, Supplemental Digital Content 11).

Meta-regression analyses were conducted to examine whether potential effect modifiers influenced treatment effects across interventions. The global tests indicated that mean age ($P = .73$) and treatment duration ($P = .58$) were not significant moderators (Tables S8 and S9, Supplemental Digital Content 12). Similarly, the regression coefficients for individual treatment comparisons were not statistically significant (all $P > .05$), suggesting that differences in mean age across studies were unlikely to influence the estimated treatment effects. TMJ diagnostic subtypes also did not significantly moderate treatment effects ($P = .12$). While a borderline significant association between low level laser and treatment effects was identified in studies involving myogenous TMD ($P = .05$), this pattern was not consistently observed across other interventions (Table S10, Supplemental Digital Content 13).

3.5. Secondary outcome

Overall, TMJ manual therapy, low-level laser, therapeutic ultrasound, and TENS showed significant improvement in MMO (TMJ manual therapy: SMD = 2.15, 95% CI = 0.64 to 3.66; low-level laser therapy: SMD = 1.65, 95% CI = 0.76 to 2.54; therapeutic ultrasound: SMD = 1.73, 95% CI = 0.47 to 2.98; TENS: SMD = 1.41, 95% CI = 0.54 to 2.28) and TMJ exercise showed a positive trend for MMO (SMD = 1.39, 95% CI = −0.06 to 2.85; Fig. 4).

The interventions were ranked based on their effect sizes on MMO as follows: TMJ manual therapy (mean rank: 1.9), therapeutic ultrasound (mean rank: 2.7), low-level laser therapy (mean rank: 3.0), TENS (mean rank: 3.6), TMJ exercise (mean rank: 3.8), and control group (mean rank: 6.0; Fig. S6, Supplemental Digital Content 14). Detailed SUCRA scores and rankings of the treatment types are provided in Table S11, Supplemental Digital Content 15. The pairwise comparison results for the study arms and individual studies are presented in Figure S7A and S7B, Supplemental Digital Content 16 and Table 3.

Sensitivity analyses using alternative pre–post correlation coefficients (0.5 and 0.3) were performed, and the network meta-analysis was repeated. The direction of SMD estimates, treatment rankings, and overall conclusions remained consistent with the primary analysis (Fig. S8, Supplemental Digital Content 17 and Table S12 and S13, Supplemental Digital Content 18).

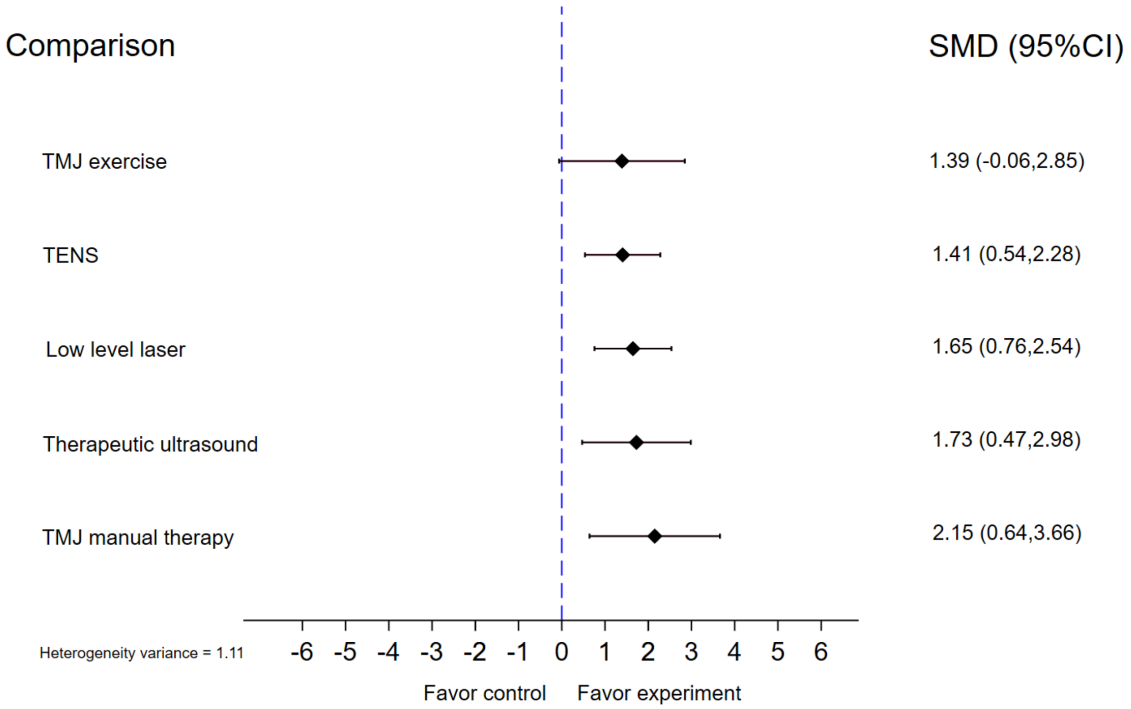


Figure 4. Forest plot of maximal mouth opening comparing interventions with control. SMD = standardized mean difference, TENS = transcutaneous electrical nerve stimulation, TMJ = temporomandibular joint.

Table 3

League table of maximal mouth opening.

Control	1.65 (0.76 to 2.54)	1.41 (0.54 to 2.28)	1.39 (−0.06 to 2.85)	2.15 (0.64 to 3.66)	1.73 (0.47 to 2.98)
−1.65 (−2.54 to −0.76)	Low level laser	−0.24 (−1.19 to 0.72)	−0.25 (−1.63 to 1.13)	0.50 (−0.85 to 1.86)	0.08 (−1.18 to 1.34)
−1.41 (−2.28 to −0.54)	0.24 (−0.72 to 1.19)	TENS	−0.02 (−1.59 to 1.56)	0.74 (−0.85 to 2.33)	0.32 (−0.94 to 1.58)
−1.39 (−2.85 to 0.06)	0.25 (−1.13 to 1.63)	0.02 (−1.56 to 1.59)	TMJ exercise	0.76 (−0.44 to 1.96)	0.34 (−1.45 to 2.12)
−2.15 (−3.66 to −0.64)	−0.50 (−1.86 to 0.85)	−0.74 (−2.33 to 0.85)	−0.76 (−1.96 to 0.44)	TMJ manual therapy	−0.42 (−2.22 to 1.38)
−1.73 (−2.98 to −0.47)	−0.08 (−1.34 to 1.18)	−0.32 (−1.58 to 0.94)	−0.34 (−2.12 to 1.45)	0.42 (−1.38 to 2.22)	Therapeutic ultrasound

TENS = transcutaneous electrical nerve stimulation, TMJ = temporomandibular joint.

Meta-regression analyses were performed to assess whether potential effect modifiers affected treatment effects across interventions. The global tests showed that participant age ($P = .22$; Table S14, Supplemental Digital Content 19), treatment duration ($P = .45$; Table S15, Supplemental Digital Content 20), and TMJ diagnostic subtypes ($P = .83$; Table S16, Supplemental Digital Content 21) were not significant moderators. Consistent with these findings, none of the regression coefficients for individual treatment comparisons reached statistical significance (all $P > .05$), indicating that variations in these factors across studies were unlikely to substantially influence the estimated treatment effects.

3.6. Inconsistency test

Side-splitting analyses for all comparisons revealed no P-values < .05, except for control treatment versus TENS for pain intensity ($P = .02$; Table S17, Supplemental Digital Content 22). Additionally, side-splitting analyses across all comparisons reported P-values > .05, confirming consistency for MMO (Table S18, Supplemental Digital Content 23).

3.7. Publication bias

Funnel plots for pain intensity and MMO are presented in Figure S9 and S10, Supplemental Digital Content 24. Egger regression tests were further conducted to assess potential small-study effects. The results yielded P-values < .01 for both outcomes, suggesting the presence of possible small-study effects or publication bias (Figs. S11 and S12, Supplemental Digital Content 25).

4. Discussion

The current network meta-analysis evaluated the comparative effectiveness of TMJ manual therapy, TMJ exercise, low-level laser therapy, therapeutic ultrasound, and TENS. Based on SUCRA values, the probability ranking of treatments for pain intensity was, in descending order, therapeutic ultrasound, TMJ manual therapy, low-level laser therapy, TMJ exercise, TENS, and control. For MMO, the ranking was TMJ manual therapy, therapeutic ultrasound, low-level laser therapy, TENS, TMJ exercise, and control. Among nonphysical modalities, TMJ manual therapy ranked higher than TMJ exercise in improving pain intensity and MMO. Among physical modalities, therapeutic ultrasound and low-level laser therapy outperformed TENS in both measures. The ranking for ultrasound therapy dropped from the first to the fourth position for MMO improvement compared to its effectiveness in reducing pain intensity.

To the best of our knowledge, this is the first network meta-analysis of RCTs examining the effectiveness of physical therapies for TMJ disorders. Al-Moraissi et al[3] conducted a network meta-analysis (36 RCTs), which focused on the hierarchy of intra-articular injections with platelet-rich plasma, hyaluronic acid, and corticosteroids; TMJ surgery; and physical therapy on pain intensity and MMO. However, the physical therapy or modality category included only low-level laser and TMJ manual therapies. Jing et al[48] focused on the effects of low-level laser therapies with different energy densities in TMJ disorders but excluded TMJ exercise, manual therapy, and other modalities. Li et al[11] performed a network meta-analysis of 26 RCTs comparing intra-articular drug injections, occlusal splints, and their combinations with control treatments. Their results highlighted the use of platelet-rich plasma as a potentially superior treatment option compared to other treatments owing to its anti-inflammatory, analgesic, and lubricating effects; however, the study did not examine the efficacy of physical therapy. Yao et al[10] included 153 trials in a network meta-analysis encompassing 59 interventions or combinations of interventions for TMJ disorders and analyzed pain intensity, function, and adverse events. However, the study did not address MMO or rank physical therapies as an independent treatment option. Our network meta-analysis addressed a critical gap by evaluating the effectiveness of physical therapies for TMJ disorders, specifically categorizing them into TMJ manual therapy, TMJ exercise, low-level laser therapy, therapeutic ultrasound, and TENS, focusing on pain reduction and MMO improvement.

Regarding the effectiveness of TMJ manual therapy and TMJ exercise in pain reduction and improvement of MMO, these may be explained by biomechanical and neurophysiological mechanisms. From a biomechanical perspective, manual therapy techniques potentially alleviate local ischemia, stimulate proprioceptive mechanisms, break down fibrous adhesions, and enhance synovial fluid production, thereby reducing pain intensity and improving MMO.[49] From a neurophysiological perspective, manual therapy has been proposed to produce immediate changes in central pain processing, resulting in pain inhibition.[50] In contrast, TMJ exercise aim to improve motor control by addressing dysfunctional muscle activation and movement patterns. These exercise involve retraining the neuromuscular system, which requires time to establish new movement patterns, enhance muscle coordination, and strengthen targeted muscles.[51] Unlike TMJ manual therapy, which provides immediate benefits through mechanical and neurophysiological effects, TMJ exercise depend on the body's active adaptation and motor learning processes, necessitating longer treatment durations to achieve noticeable and sustained improvements.[51] Consequently, the effectiveness of TMJ exercise may take longer to manifest than that of TMJ manual therapy. Based on this rationale, TMJ manual therapy demonstrates superior short-term effectiveness compared to TMJ exercise.

Among physical agent modalities, therapeutic ultrasound and low-level laser therapy ranked higher than TENS in reducing pain intensity and improving MMO. TENS primarily functions through the gate control theory and endogenous opioid release,[52] but its role in tissue repair remains inconclusive.[53] In contrast, therapeutic ultrasound and low-level laser therapy provide analgesic and tissue repair benefits. Therapeutic ultrasound modulates immune responses, including vasoconstriction, cell adhesion, phagocytosis, and the production of growth factors and cytokines, thereby enhancing tissue repair.[54] low-level laser therapy significantly reduces local inflammatory markers, such as PGE2 and TNFα.[55] Additionally, it enhances the proliferation of fibroblasts, keratinocytes, endothelial cells, lymphocytes, and growth factors, thereby promoting tissue repair.[56] These effects may explain why low-level laser therapy and therapeutic ultrasound outperform TENS.

Furthermore, while therapeutic ultrasound initially ranked higher than TMJ manual therapy in reducing pain intensity, the ranking of therapeutic ultrasound dropped when improvements in MMO were assessed. This difference can be explained by the principles of biomechanics and arthrokinematics. From a biomechanical perspective, tissue extensibility is enhanced by the thermal effects of therapeutic ultrasound, which reduces stiffness by altering material characteristics.[57] However, increasing tissue extensibility alone is insufficient to achieve tissue deformation or restore joint mechanics, as these processes require the application of mechanical stress.[58] Manual therapy directly applies mechanical force to the tissues.[58] From an arthrokinematic perspective, normal joint movement depends on proper accessory motions, such as rolling and gliding.[59] Manual therapy enhances these accessory motions by addressing joint restrictions and restoring optimal joint mechanics,[59] thereby improving function. By ameliorating tissue strain through biomechanics and enhancing accessory motions through arthrokinematics, manual therapy demonstrated a superior ability to increase the range of motion compared to therapeutic ultrasound.

This network meta-analysis has several limitations. First, although the meta-regression analysis indicated that TMJ diagnostic subtypes were not a major moderator, a borderline trend toward statistical significance was observed for pain outcomes (TMJ disorder-myogenous). Future research should further investigate and differentiate specific TMJ disorder subtypes to better clarify their potential influence on treatment effects. Second, treatment duration varied across studies, making the comparison of outcomes difficult. Further studies are required to clarify the relative benefits of short- and long-term treatment strategies. Third, this analysis focused only on short-term posttreatment effects. Incorporating long-term data into future research could provide a more comprehensive assessment. Fourth, most included studies did not clearly specify whether MMO was measured as active or passive mouth opening. This lack of clarity may introduce methodological heterogeneity and affect the comparability of results across studies. Future studies should clearly report the measurement procedures for MMO to improve the consistency and interpretability of findings. Fifth, studies involving combination therapies were excluded to ensure that each node represented a single treatment category. However, TMJ disorders rehabilitation commonly involves multimodal treatment strategies in clinical practice. As a result, this approach may reduce the ecological validity of the findings and limit the available evidence base. Future studies should investigate the effectiveness of combined therapeutic approaches to better reflect real-world clinical practice. Sixth, the treatment categories used in this study may be relatively broad. For example, manual therapy, exercise, and therapeutic ultrasound may include different techniques or parameters, which could introduce clinical heterogeneity within treatment categories. Future studies should further stratify specific treatment techniques and parameters to provide more precise estimates of treatment effects. Seventh, although SUCRA values provide a useful summary of the probability ranking of treatments, they reflect the likelihood of being among the best interventions rather than the magnitude of treatment effects. Therefore, treatment rankings should not be interpreted as definitive evidence of clinical superiority. For example, although therapeutic ultrasound and TMJ manual therapy showed relatively higher SUCRA rankings for pain intensity, the differences were small and should be interpreted cautiously. Similarly, TMJ exercise and TENS demonstrated relatively favorable rankings for MMO, but these differences were also modest. It should not be interpreted as definitive evidence of clinical superiority. Furthermore, the pairwise comparisons (Fig. S4, Supplemental Digital Content 8 and Fig. S7, Supplemental Digital Content 16) between therapeutic ultrasound and TMJ manual therapy for pain intensity, as well as between TMJ exercise and TENS for MMO, showed small effect sizes and wide CIs, indicating considerable uncertainty and limited precision in the estimated treatment effects. Therefore, the ranking results should be interpreted cautiously. Finally, side-splitting analyses revealed significant inconsistencies between direct and indirect evidence in the comparison of control versus TENS for pain intensity. These discrepancies underscore the challenges in synthesizing evidence across studies. Future research should aim to reduce heterogeneity by standardizing intervention protocols and improving study designs.

5. Conclusions

This network meta-analysis suggests that TMJ manual therapy had a higher probability ranking than TMJ exercise for pain reduction and improvement in MMO. Among physical modalities, therapeutic ultrasound and low-level laser therapy tended to rank higher than TENS. From a clinical perspective, reductions in pain intensity may improve comfort during jaw movement and daily oral activities, while increases in MMO may facilitate functional activities such as chewing and speaking. In terms of treatment selection, TMJ manual therapy may represent a practical first-line option for TMJ disorders, whereas physical modalities such as therapeutic ultrasound and low-level laser therapy may serve as adjunctive treatments when available, with treatment decisions. However, the limitations include inconsistent treatment durations and a lack of data on TMJ disorder subtypes, underscoring the need for further research that should focus on long-term outcomes in different subtypes of TMJ disorders.

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